

Advanced Python Programming for Data Science

CH – 05

Data Visualization and Storytelling

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Data Visualization

- ✓ Data visualization is the graphical representation of information and data. It transforms complex datasets into visual formats like charts, graphs, and maps, enabling users to identify patterns, trends, and insights quickly and effectively.

- **Storytelling with Data:** Visuals transform raw data into compelling narratives.

- ✓ **Importance of Data Visualization**

- **Simplifies Complex Data:** Converts large datasets into easy-to-understand visuals.
 - **Supports Decision-Making:** Provides actionable insights for stakeholders.
 - **Enhances Communication:** Facilitates clear storytelling with data.
 - **Identifies Patterns and Trends:** Unveils hidden relationships in the data.

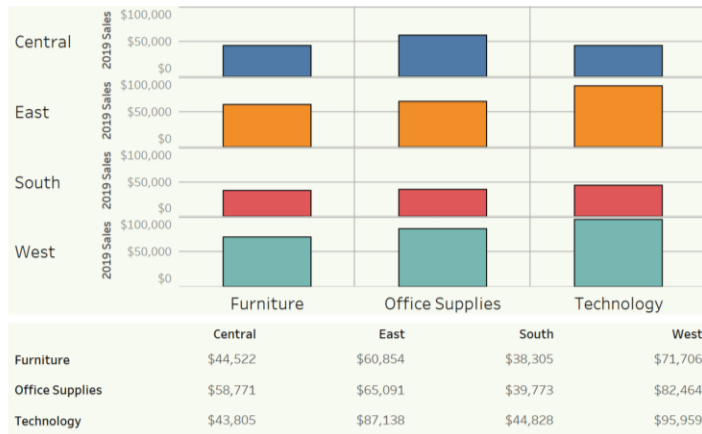
- ✓ **Example:**

- **Real-Time Dashboards:** Visualizations in dashboards help track metrics and KPIs continuously.

Principles of Effective Data Visualization

Compare These Visuals!

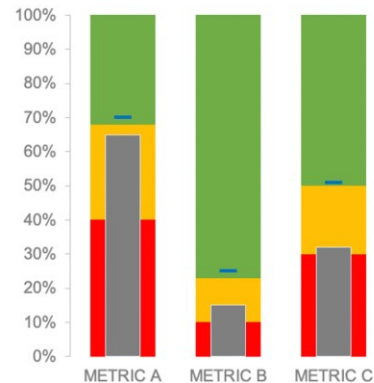
sales by region and category



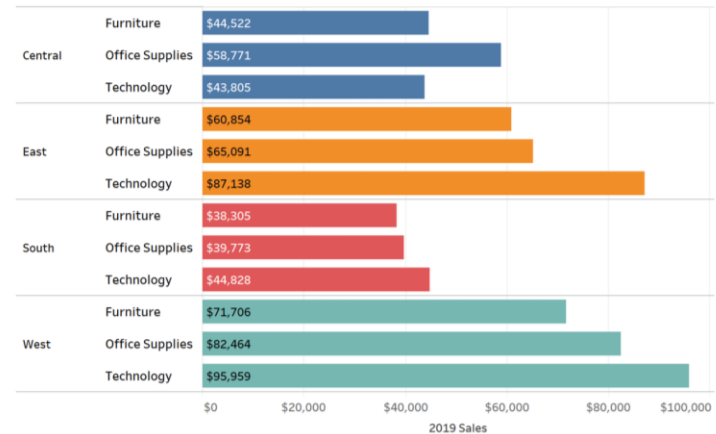
! ineffective

OVERWHELMING COLOR

Performance metrics



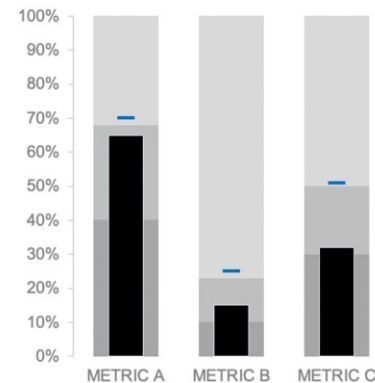
sales by region and category



✓ effective

A BETTER ALTERNATIVE

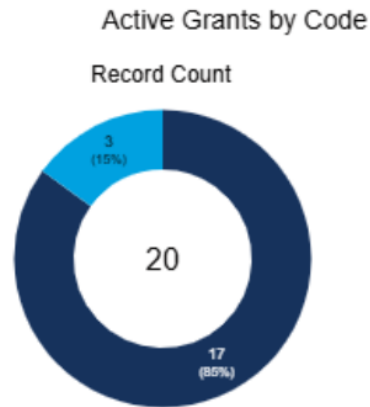
Performance metrics



Principles of Effective Data Visualization

Compare These Visuals!

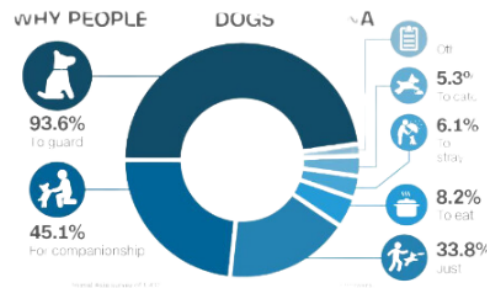
The Good



- Its simple and has color unity.
- Correct Chart type for the data

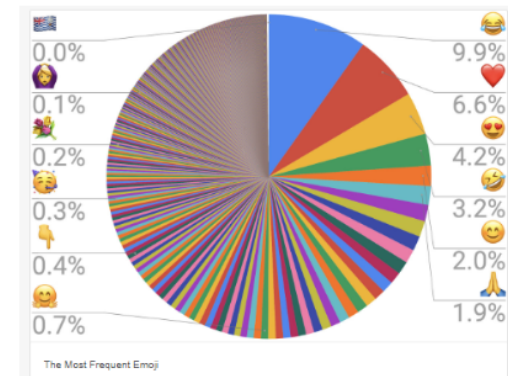


The Bad



- Has color unity.
- Too Many categories for a pie chart.
- In a Pie Chart, the whole should add to 100%, here the percentages go up to 195% because of the use of multipicklists in the measures.

The Ugly



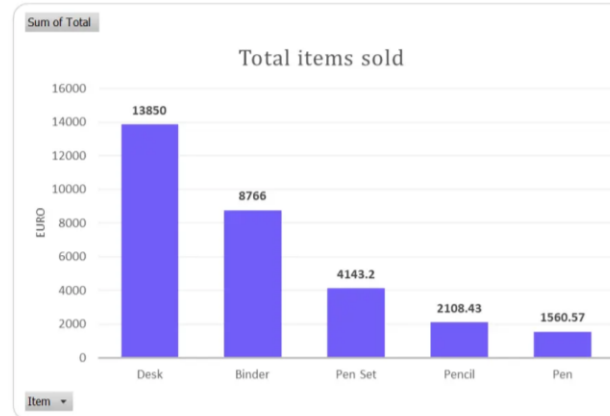
- No color unity.
- Incorrect chart type for the data.
- Information Overload, leads to lack of focus.
- = You lost the audience.

Principles of Effective Data Visualization

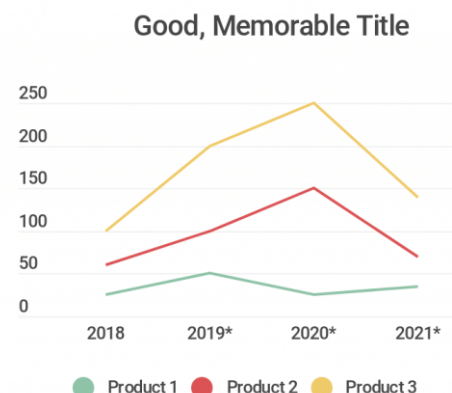
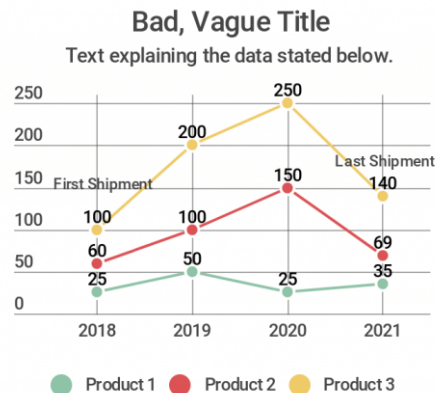
Compare These Visuals!



Cluttered chart
(bad data visualization example)

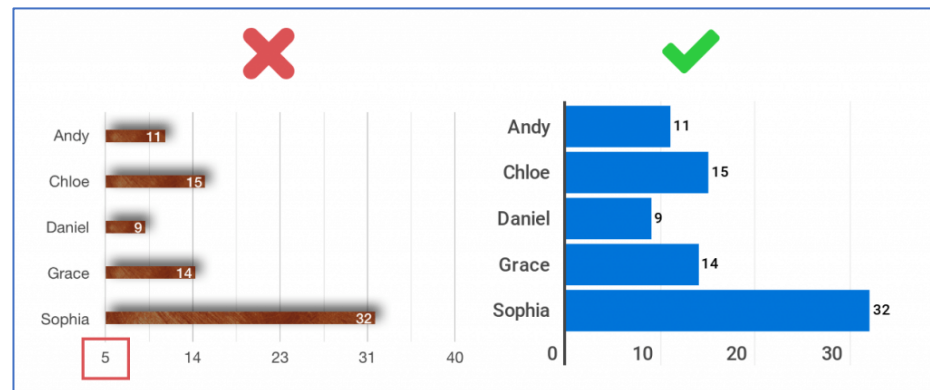
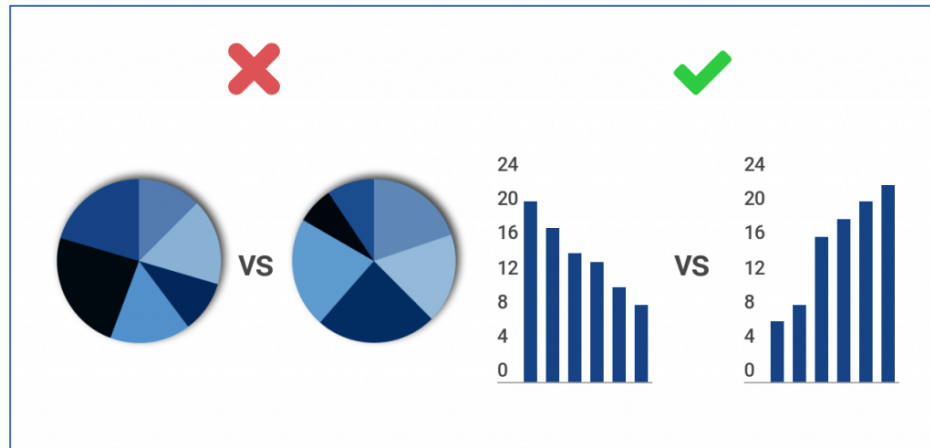


Clean chart
(good data visualization example)



Principles of Effective Data Visualization

Compare These Visuals!



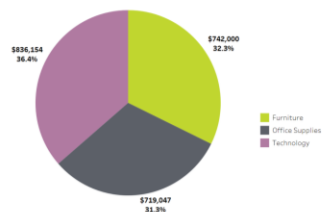
Principles of Effective Data Visualization

- ✓ Misleading visualizations distort data insights, leading to incorrect interpretations.
 - Effective visualization principles ensure **clarity, accuracy, and unbiased communication**.

- ✓ **Clarity:** Use simple, clear designs that emphasize the data's message. Avoid unnecessary decorations (*chartjunk*).
 - **Readable Text:** Use legible font sizes and clear labels for axes, legends, and titles.
 - **Focus on the Data:** Avoid visual distractions like excessive colors, 3D effects, or unnecessary elements.
 - **Highlight Key Insights:** Use emphasis, such as color or annotations, to guide the audience to the main message.

Principles of Effective Data Visualization (Cont..)

- ✓ **1. Accuracy:** Maintain proportionality in data representation. Misleading scales, truncated axes, or skewed visuals can exaggerate trends.
- ✓ It reflects the truthful representation of data without distortions or manipulations.
 - **Proportional Representations:** Ensure that visual elements like bar lengths or pie slice sizes match the data values.
 - **Avoid Truncated Axes:** Starting axes at non-zero values can mislead viewers by exaggerating trends.
 - **Appropriate Charts and Scales:** Use consistent scales to represent the data correctly.



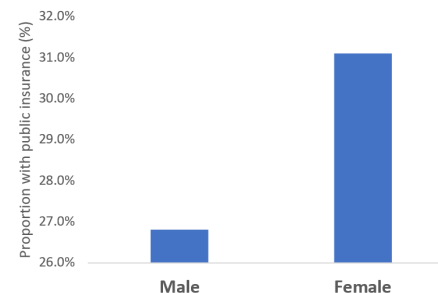
! ineffective

sales by product category

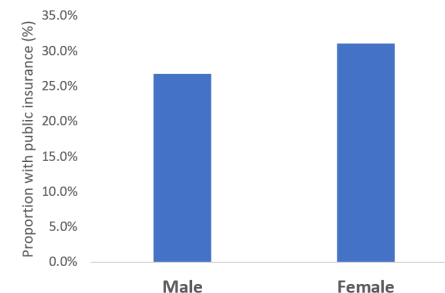


✓ effective

Panel A



Panel B

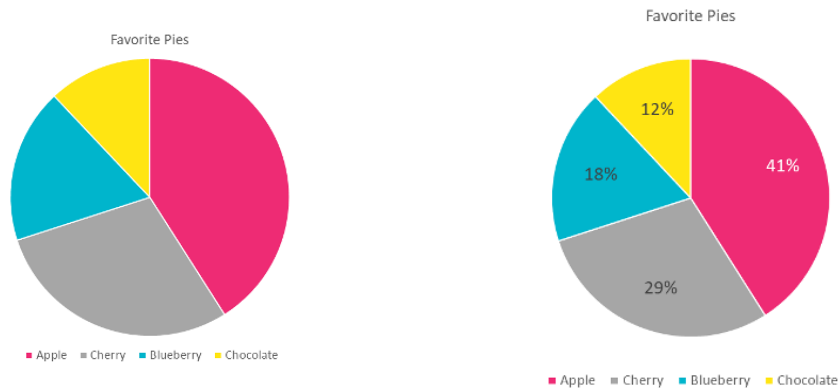


Principles of Effective Data Visualization

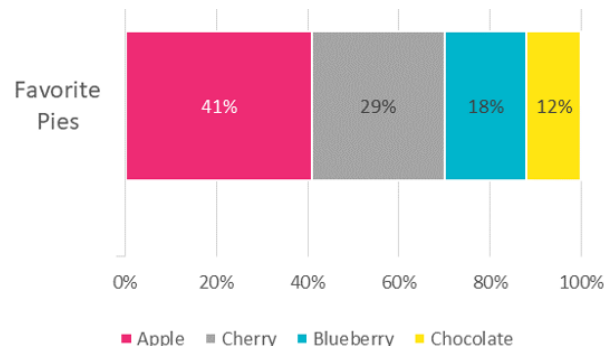
Accuracy in Interpreting Data Visualizations

✓ **Choose right chart:** E.g.: Avoiding Pie/Donut Charts

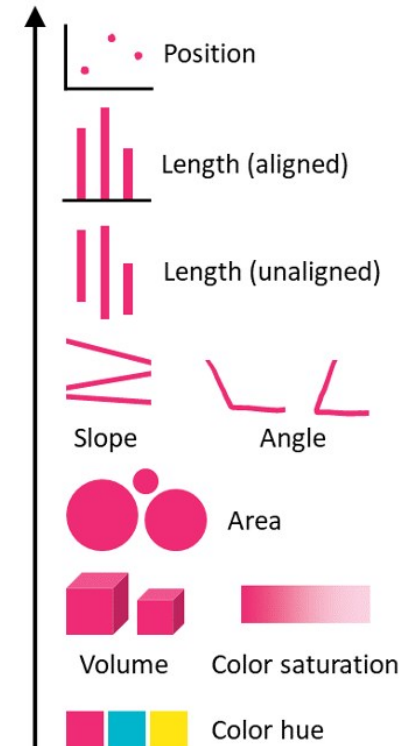
- In general, it is best to use a pie chart or donut chart only if you are representing two categories where one is larger than the other.



- If you have more than two categories, consider using an approach, such as a horizontal stacked bar chart.



Easier to accurately interpret



Harder to accurately interpret

Principles of Effective Data Visualization (Cont..)

- ✓ **2. Consistency:** Consistency ensures uniformity in the design to avoid misinterpretation and promote understanding.
 - **Standardized Scales and Units:** Use the same scales, units, and baselines across multiple charts for comparisons.
 - **Unified Style:** Maintain consistent colors, fonts, and chart types within a dataset.
 - **Repetition for Patterns:** Consistency helps establish recognizable patterns for the audience.

- ✓ **3. Simplicity:** Simplicity removes unnecessary complexity to make the visualization accessible to a broader audience.
 - **Minimal Design:** Include only elements essential to the story the data tells.
 - **Avoid Overloading:** Present limited data per visualization to prevent overwhelming the viewer.
 - **Streamlined Layout:** Use clean layouts with for better readability.

Principles of Effective Data Visualization (Cont..)

✓ Best Practices for Data Visualization in EDA

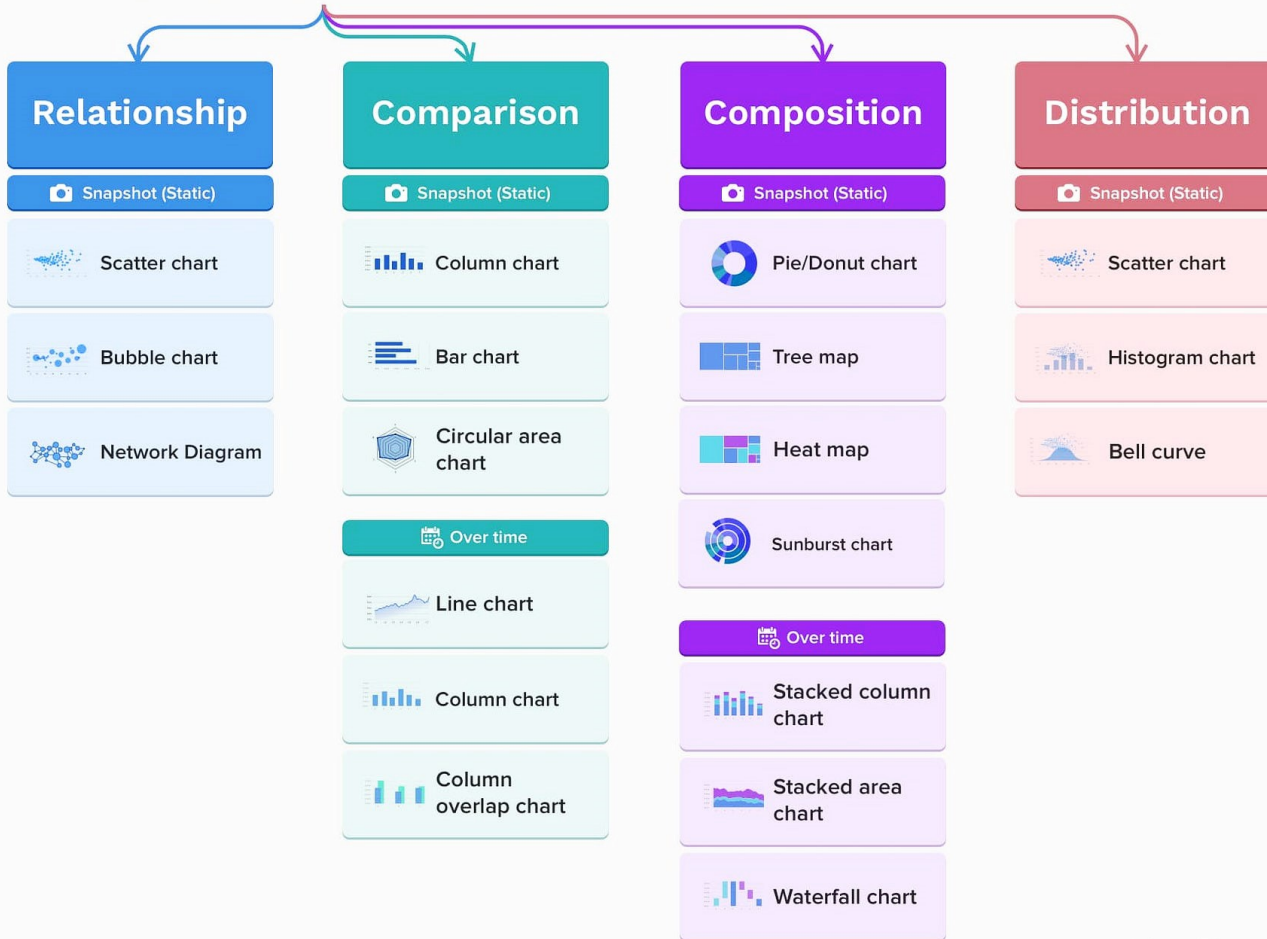
- 1. Choose the Right Chart Type:** Match the visualization to the data type and analysis goal.
- 2. Simplify Visuals:** Avoid unnecessary clutter; keep visuals focused and clean.
- 3. Highlight Key Insights:** Use color, annotations, and labels to emphasize important patterns.
- 4. Ensure Accuracy:** Avoid misleading scales, distorted proportions, or incomplete data.
- 5. Keep Consistent Design:** Use the same font size, colors, and axis labeling style across all charts to make them easier to interpret.
- 6. Provide Context:** Include titles, legends, and clear axis labels to help viewers understand the visualizations without confusion.

Principles of Effective Data Visualization (Cont..)

Choosing Right Visualization

👉 Choose the right chart

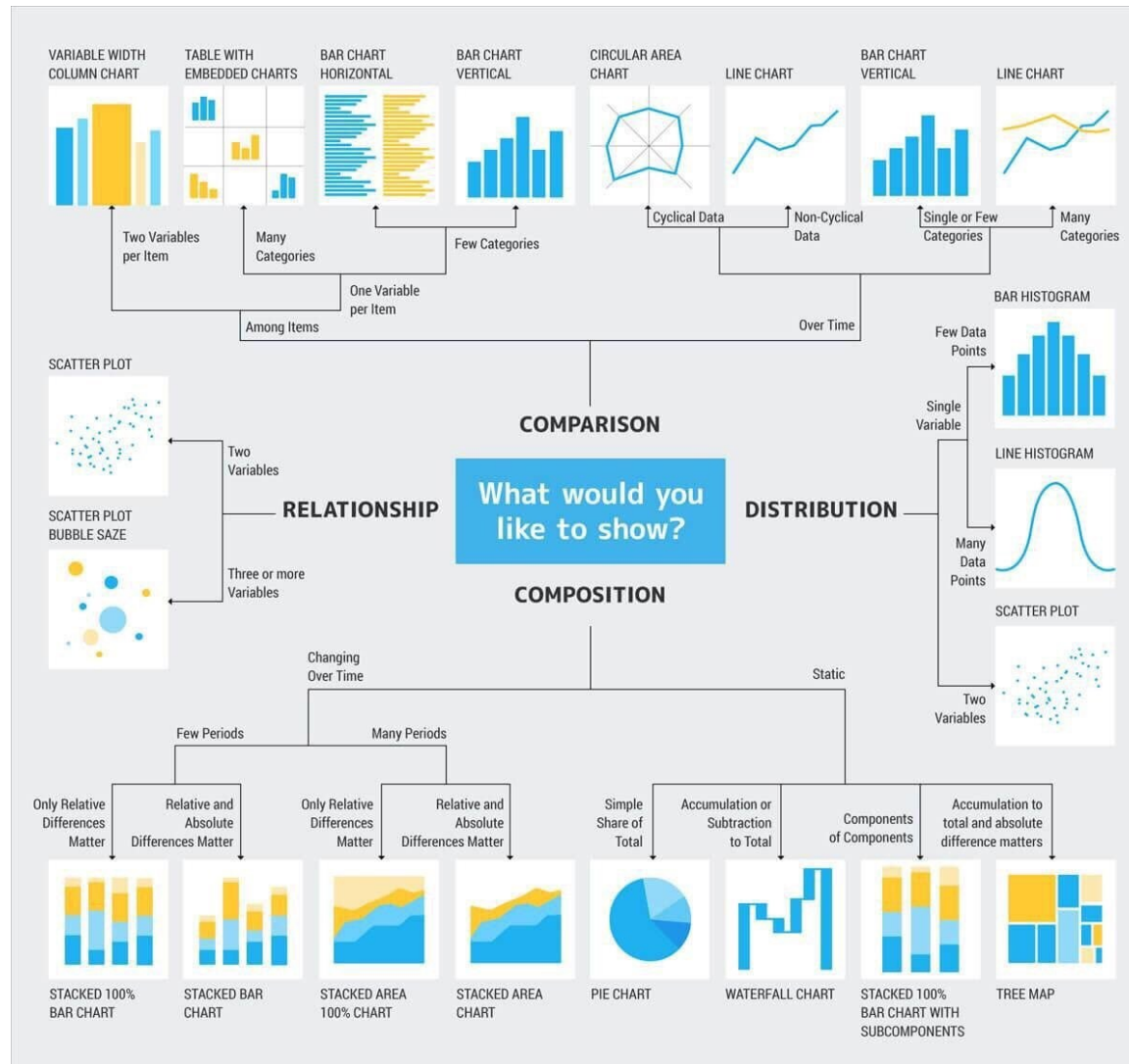
I would like to **see**



by Taras Bakusevych

Principles of Effective Data Visualization (Cont..)

Choosing Right Visualization



Data Visualization Types (Cont..)

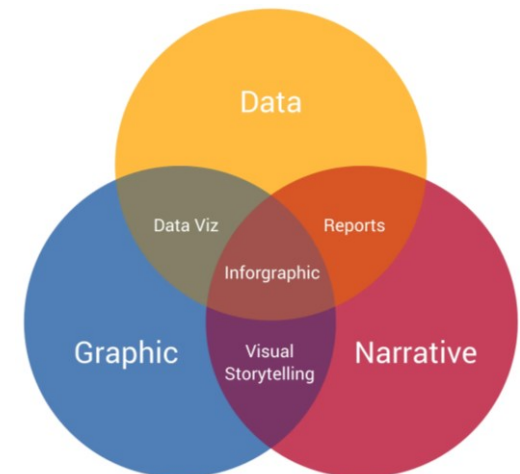
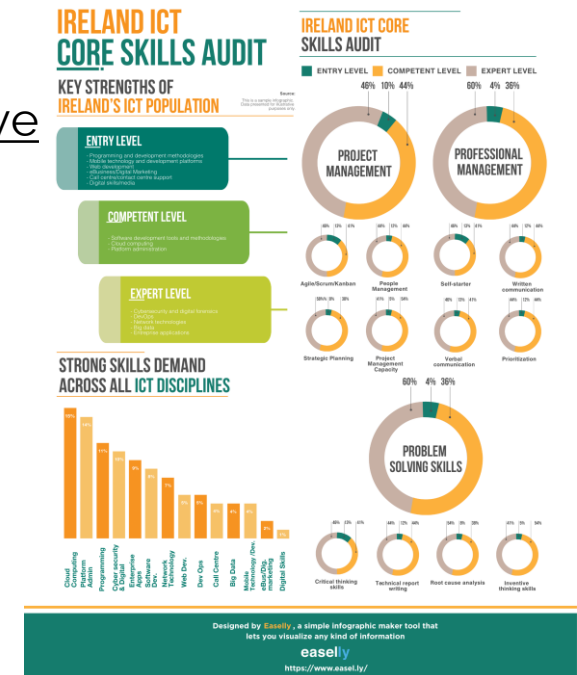
Infographics and Visualization

✓ **Infographics:** Infographics combine visuals, narrative text, and graphic design to communicate a specific story or concept in a visually appealing way.

- It often **contains multiple data visualizations**
- **Purpose:** Focuses on storytelling, often incorporating creative elements and non-quantitative data.
- **Examples:** Newspapers Poster, Educational posters.

✓ **Data Visualization:** Data visualization focuses on accurately representing numerical data using visual elements for analysis and decision-making.

- **Purpose:** Provides an accurate and efficient way to explore and analyze quantitative data.
- **Examples:** Bar charts, line charts, donut chart



Data Visualization

Practical Applications

✓ Business

- **Sales Performance:** Bullet charts and line charts track sales trends and targets.
- **Customer Segmentation:** Scatter plots and bubble charts identify customer groups.

✓ Healthcare

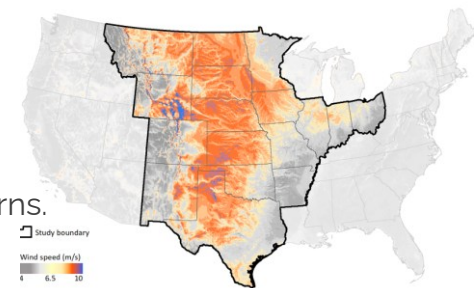
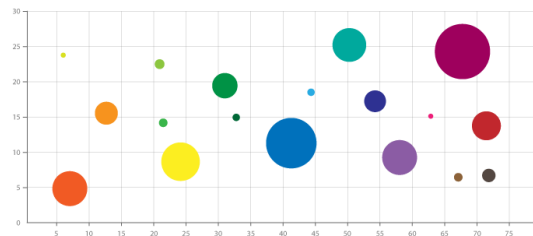
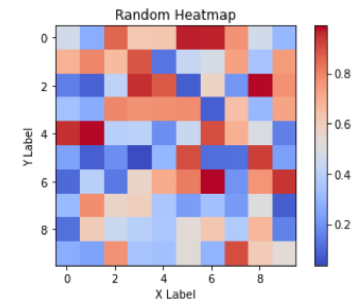
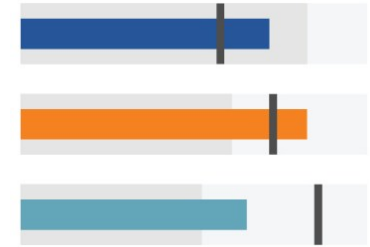
- **Patient Data:** Box plots and histograms show test result variations.
- **Disease Spread:** Geospatial maps and time-series line charts monitor trends.

✓ Education

- **Student Scores:** Histograms and box plots show performance distributions.
- **Attendance Trends:** Line and bar charts track attendance over time.
- **Curriculum Effectiveness:** Heatmaps identify correlations between methods and outcomes.

✓ Finance

- **Portfolio Analysis:** Scatter and bubble charts compare investment risks and returns.
- **Expense Monitoring:** Pie charts and treemaps visualize spending categories.



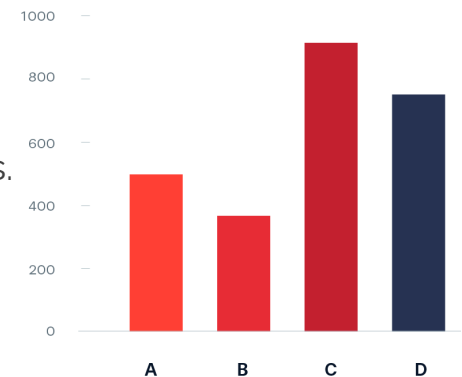
Data Visualization Techniques

- ✓ Data visualization techniques use charts, plots, and dashboards to represent data in a clear, insightful manner, making it easier to identify patterns, trends, and relationships.



- ✓ **Charts:**

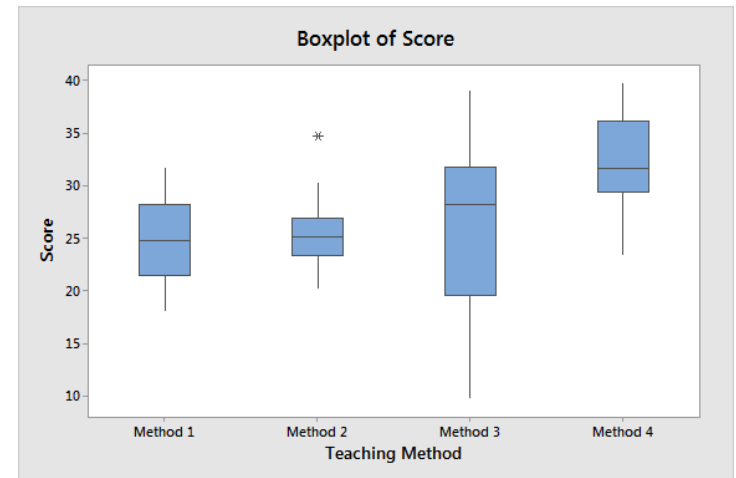
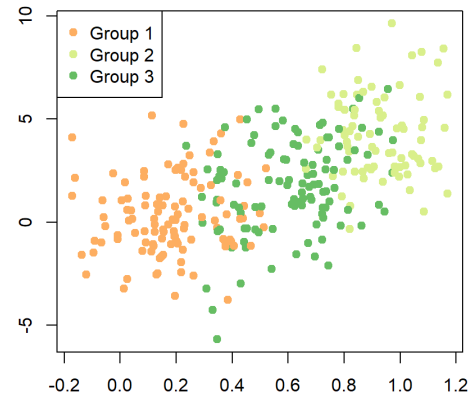
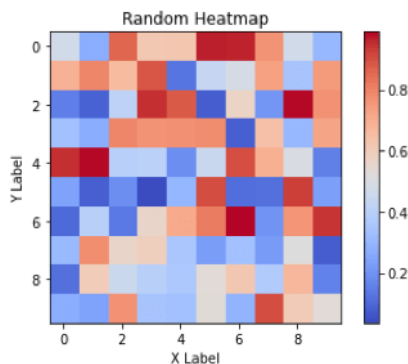
- A chart is a graphical representation of data to simplify understanding and identify patterns, trends, or insights.
- Typically summaries trends, comparisons, or summaries for non-technical audiences in presentations, reports, or general communication.
- Key Characteristics:
 - Focus on simplicity and clarity for quick interpretation.
 - Effective for conveying high-level insights to broad audiences.
- Focus: Charts target non-technical users
- Examples: Bar, Line, Pie Charts



Data Visualization Techniques (Cont..)

✓ Plots

- Plots are specialized visualizations used in technical and scientific contexts to explore detailed data patterns, distributions, or relationships, aiding analysts and researchers in extracting deeper insights.
- Key Characteristics:
 - Focus on depth and precision in representing data.
 - Suitable for multivariate, distributional, or correlation analysis.
 - Common in technical reports, research papers, or exploratory data analysis.
- Focus: Plots are suitable analysts and researchers
- Examples: Scatter, Box Plot, Heatmap



Data Visualization Techniques (Cont..)

✓ Dashboards

- Dashboards are **interactive platforms that consolidate multiple visualizations**, metrics, and data sources into a single interface, offering real-time monitoring and decision-making tools for business or operational contexts.
- Key Characteristics:
 - Integrates diverse charts, plots, and KPIs into one view.
 - Allows user interaction, such as filtering or drilling down for details.
 - Frequently used for strategic decision-making and operational monitoring.
- Focus: dashboards are suitable for decision-makers.
- Examples: Sales Dashboard, Website Analytics, Stock Monitoring Dashboard



Tools for Data Visualization

- ✓ **Matplotlib:** A Python library for creating static, interactive, and animated plots. It provides fine control over chart appearance but requires more coding for advanced plots.
 - **Customization:** Fine control over plot elements.
 - **Versatility:** Supports static, animated, and interactive visualizations.

- ✓ **Seaborn:** Built on top of Matplotlib, Seaborn simplifies the creation of attractive, informative statistical graphics with built-in themes and color palettes.
 - **Ease of Use:** Simplifies complex visualizations with minimal code.
 - **Aesthetics:** Attractive default themes and color palettes.

Data Visualization Techniques (Cont..)

- ✓ **Tableau:** A powerful tool for creating interactive and shareable dashboards with a user-friendly drag-and-drop interface.
 - **Interactivity:** Create dynamic, interactive dashboards.
 - **Ease of Use:** Drag-and-drop interface for quick visualizations.
- ✓ **Power BI:** A Microsoft tool for data analysis and visualization, seamlessly integrating with Microsoft services to build robust dashboards and reports.
 - **Integration:** Seamlessly connects with various data sources, especially Microsoft products.
 - **Advanced Analytics:** Built-in AI for deeper insights.
- ✓ **Google Data Studio:** A free tool for creating interactive reports and dashboards, ideal for those using Google services.
 - **Integration:** Connects easily with Google Analytics, Ads, and other Google products.
 - **Collaboration:** Enables real-time collaboration and easy sharing of reports.

Visualization with Matplotlib

- ✓ Creating Line, bar, histogram, scatter, subplots.

Data Visualization Techniques (Cont..)

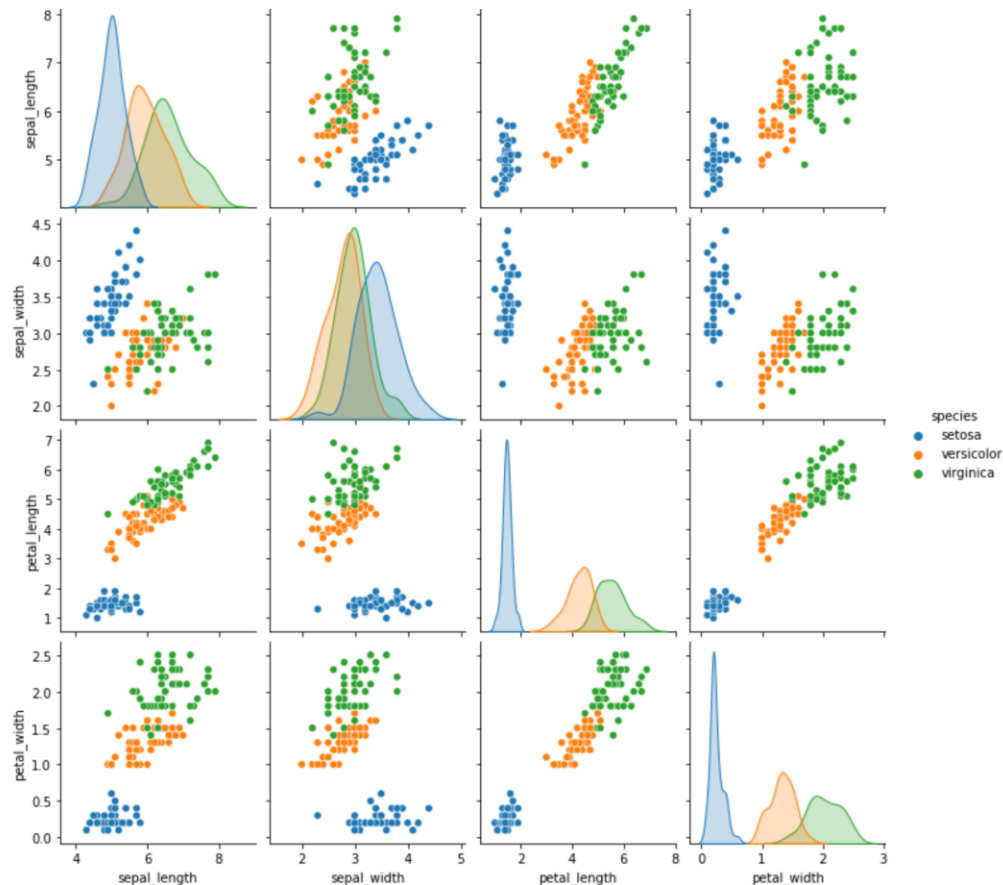
Seaborn for statistical visualization

- ✓ Box plot, pair plot, heat map

Data Visualization Techniques (Cont..)

Pair Plot

- ✓ **Non-diagonal positions:** Scatter Plots of corresponding features
- ✓ **Diagonal positions:** Distribution Plots



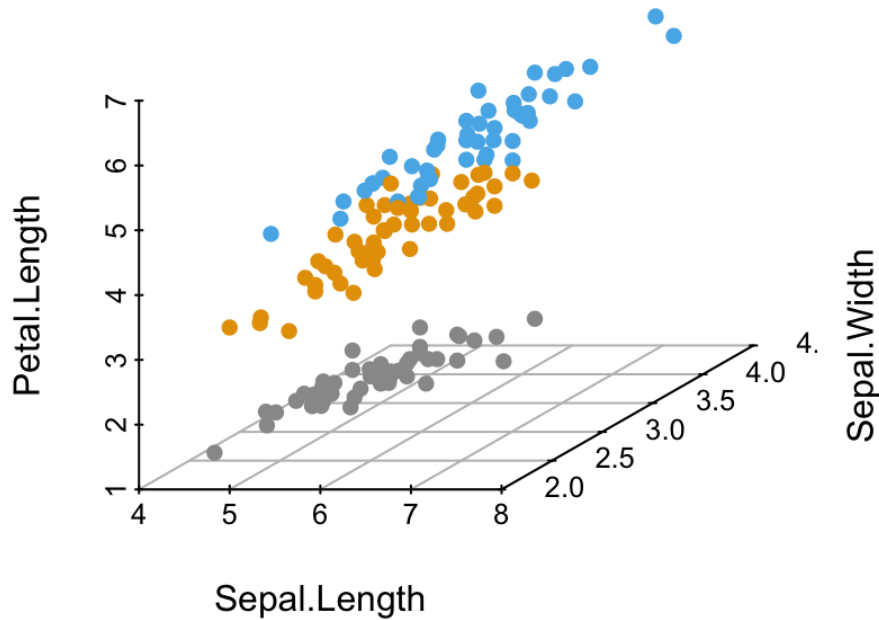
Interactive visualization using Plotly

- ✓ Student presentation!

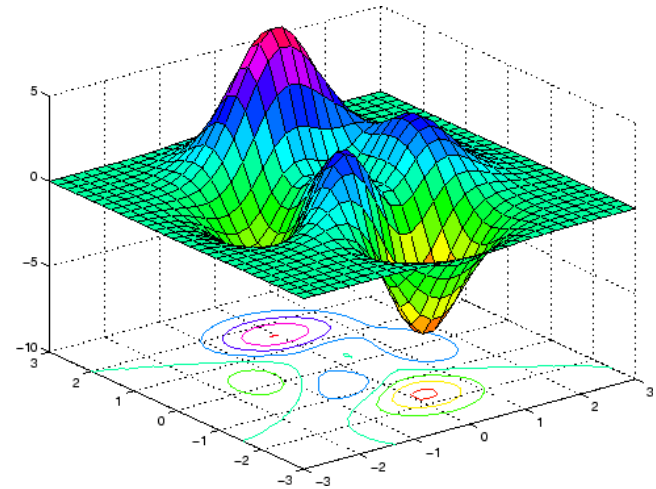
Data Visualization Techniques (Cont..)

3D plots

✓ 3D Scatter Plot



Surface Plot



Visualization driven insight generation

✓ Student Presentation!

✓ Student Presentation!